# Practical 6

## Problem Definition

Design a 4-bit synchronous counter that counts from 0000 (0) to 1001 (9) and then resets automatically to 0000. Use T Flip-Flops for implementation. The counter should operate in a synchronous manner, meaning all flip-flops are triggered by the same clock pulse. The design must include:

1. State diagram representing the 10 valid states (0–9).

2. Excitation table showing required T inputs for each flip-flop transition.

3. K-map minimization to derive simplified Boolean expressions for each T input.

4. Implementation and verification through simulation, ensuring the counter resets after 9 (MOD-10 operation).

## Requirements / Specifications:

●Use four T Flip-Flops ($T_3$, $T_2$, $T_1$, $T_0$).

●The counter must follow the sequence 0000 → 0001 → 0010 → … → 1001 → 0000.

●Include an automatic reset logic for invalid states (1010–1111).

●Derive excitation table and minimized Boolean equations for all T inputs.

●Verify operation via simulation and truth table. Tools/Technology To Be Used

●Simulation Software: Logisim Evolution / Proteus / Multisim / Tinkercad Circuits

●Components: 4 × T Flip-Flops, AND/OR gates for reset logic

●Verification Method: Truth table and waveform observation

## Aim :

To design and implement a 4-bit synchronous MOD-10 counter using T Flip-Flops that counts from binary 0000 (0) to 1001 (9) and automatically resets to 0000, ensuring correct synchronous operation and verification through simulation.

## T flip flop:

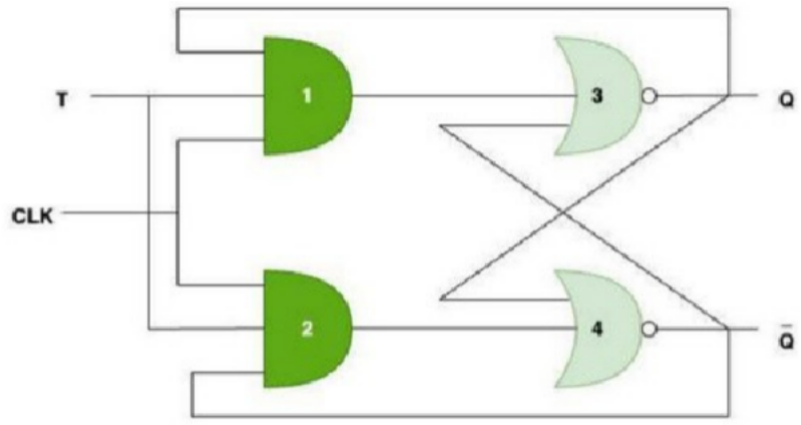
## Truth table:

| T | Q | Q(t+1) |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |

## Excitation Table:

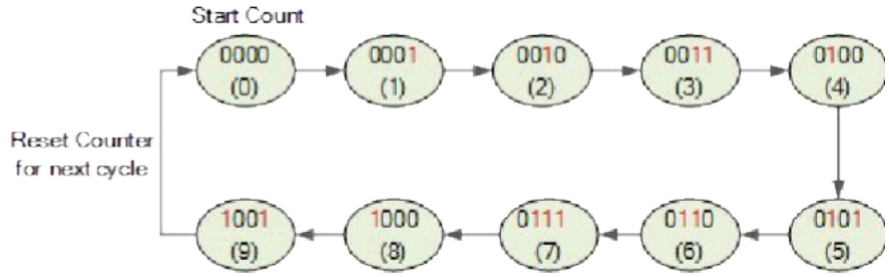
| $Q_t$ | Q(t+1) | T |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |
| 0 | 1 | 1 |

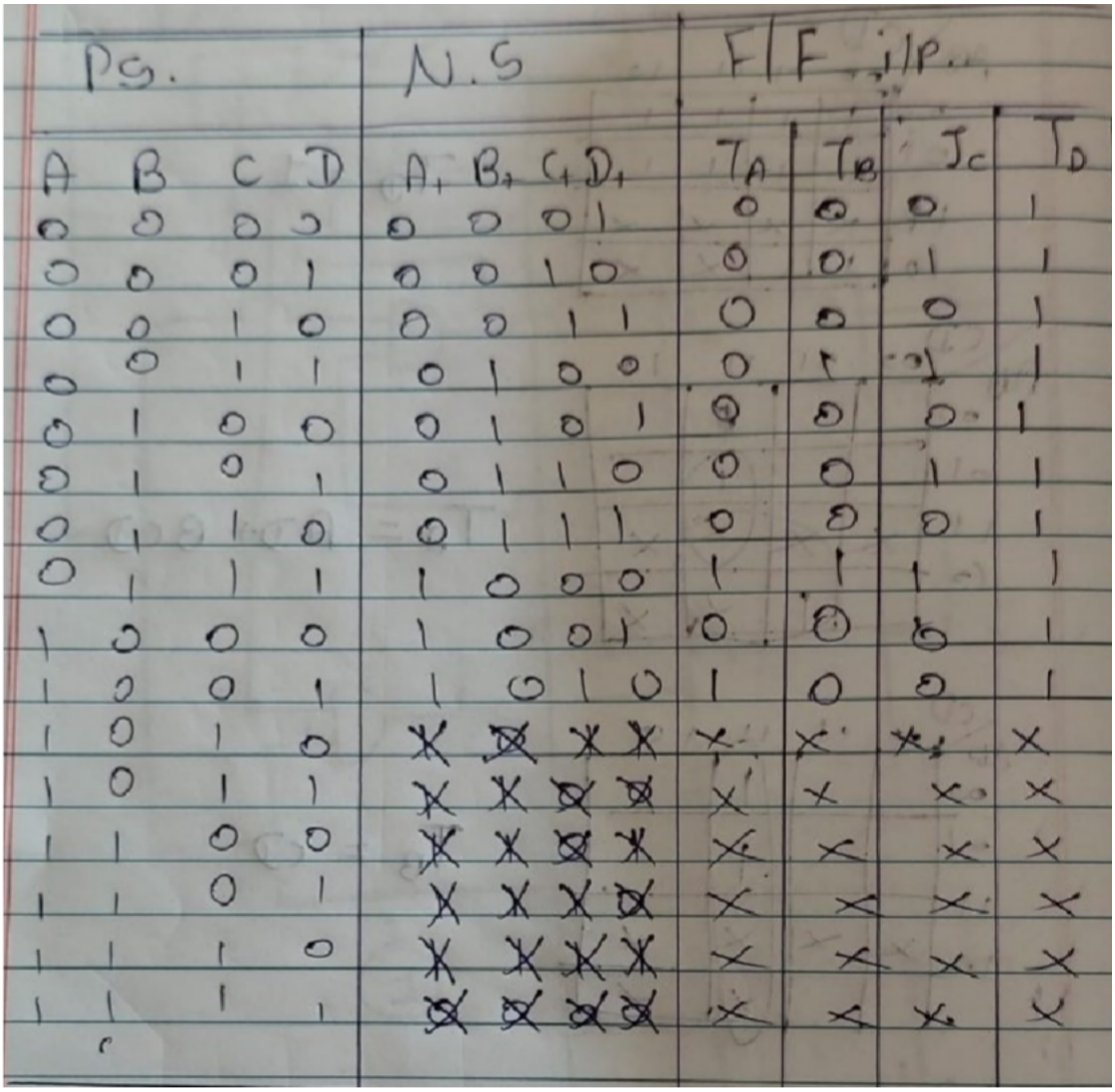
## Circuit Diagram:



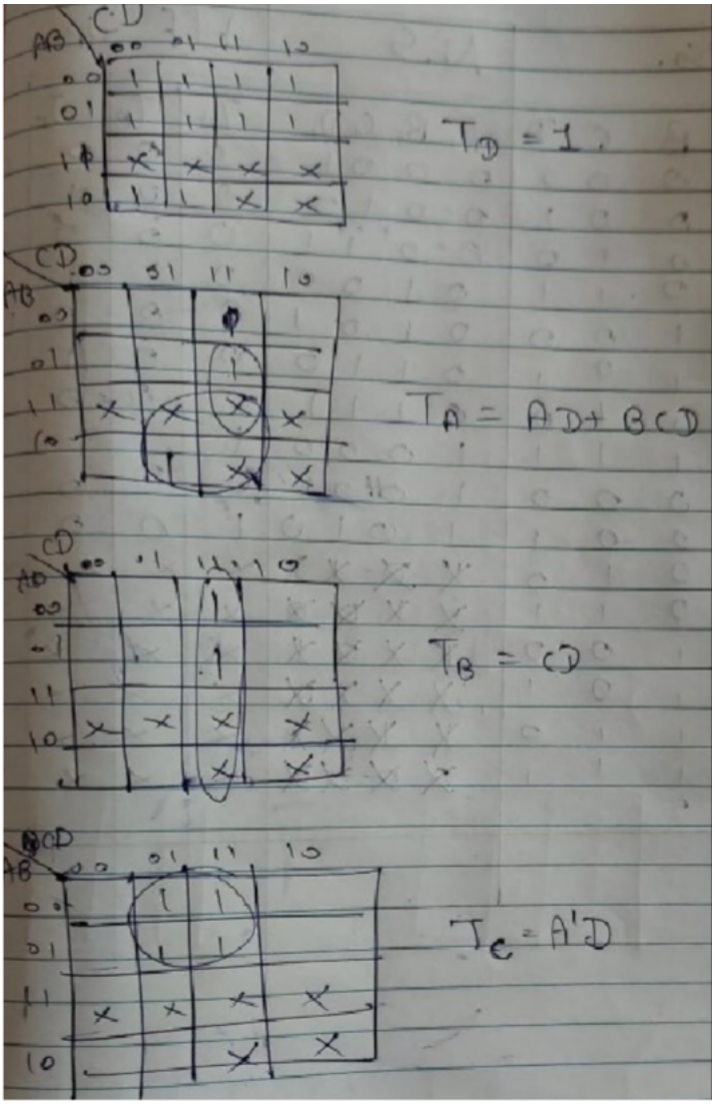
## 4-bit binary 0-9 counter:

### State diagram & state table:

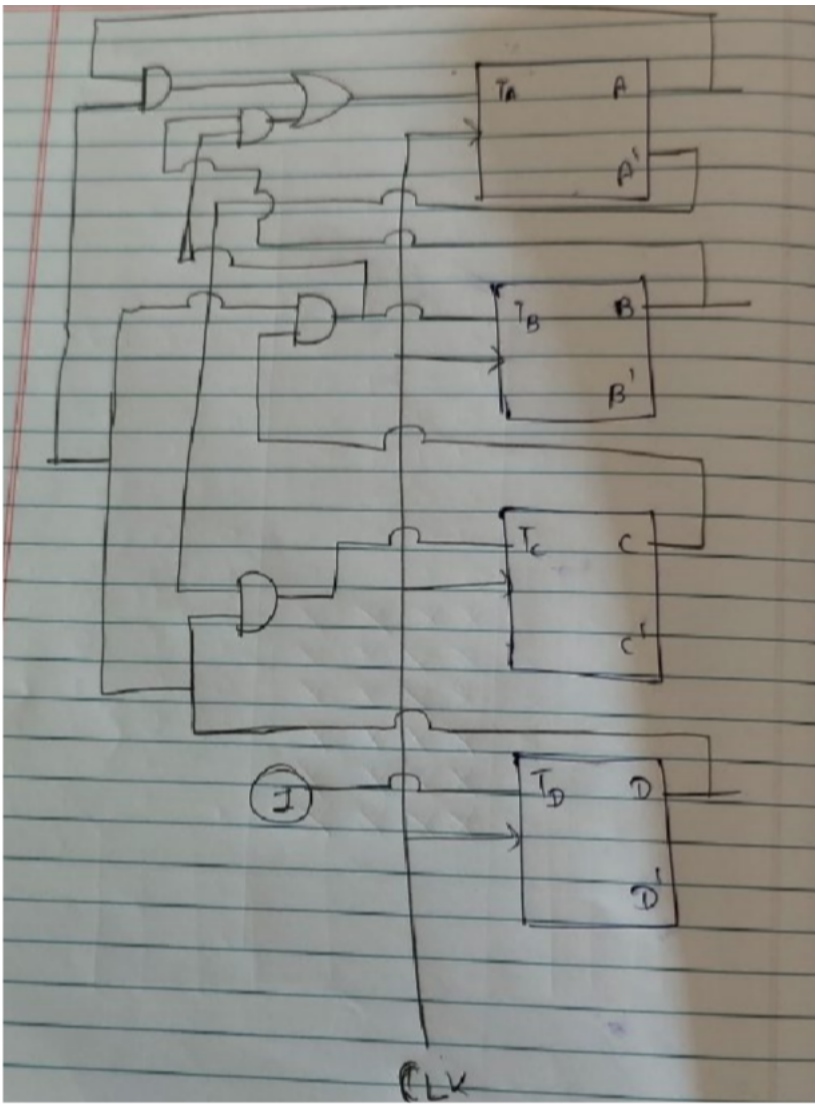


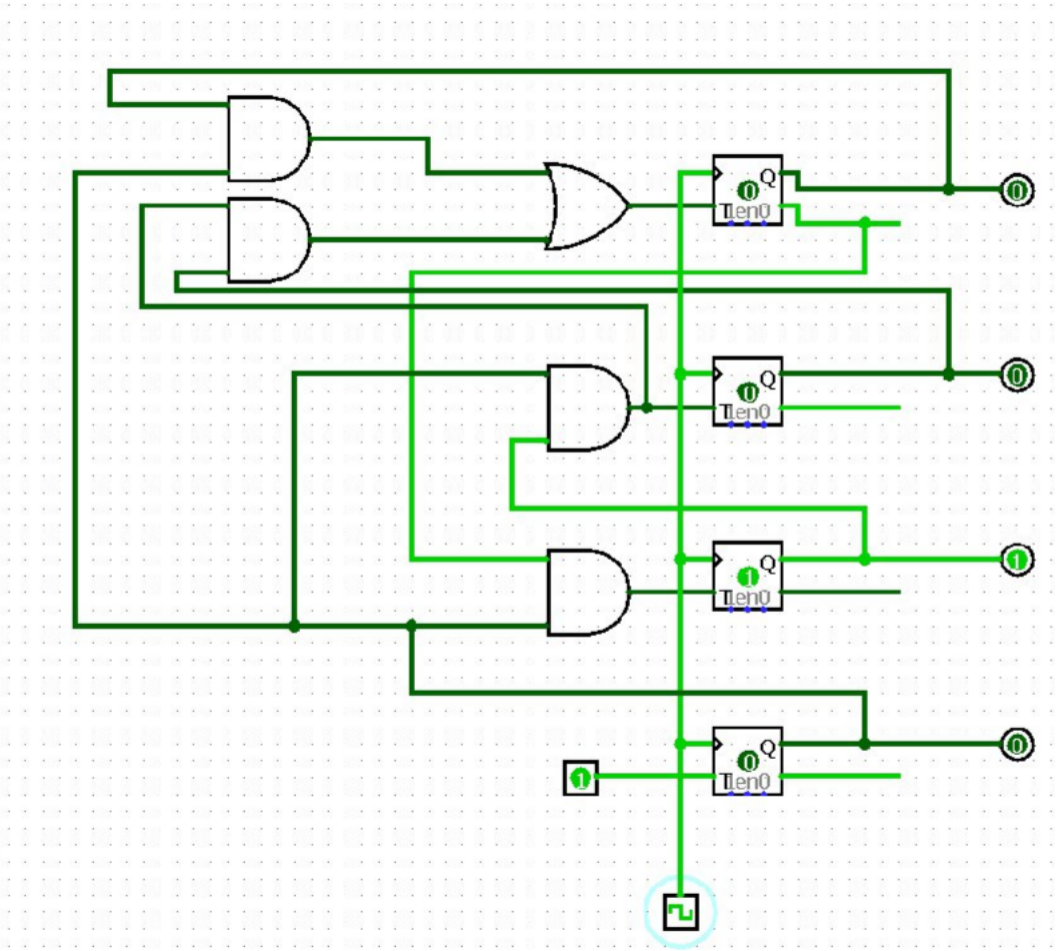
**K-Map:**



**Circuit Diagram:**



***Screen shot:***




**Conclusion:** This practical successfully demonstrates the design of a 4-bit bidirectional synchronous counter utilizing T flip-flops to achieve a toggle-based counting sequence. By applying K-map simplification to the excitation tables, the design proves that a single MODE control signal can effectively manipulate the toggle logic to switch between up and down counting modes. The inclusion of synchronous LOAD and asynchronous CLEAR functions highlights the precision of clock-driven data entry versus the immediate priority of hardware resets. Ultimately, the simulation confirms that using T flip-flops provides a streamlined and hardware-efficient approach to building versatile sequential circuits with synchronized state transitions.